UNIVERSIDAD LIBRE · SYSTEMS ENGINEERING · SECURITY POSTURE: HIGH LEVEL

Farmaps

Cybersecurity and service management with an ITIL approach

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Cybersecurity & Ethical Hacking · 2026 · BOGOTÁ D.C.

4 Modules	P1 – P4	S17 – S22
P06 · P07 · A07 · A08	ITIL SLA Matrix	Ethical Hacking

01 The project

CONTEXT & SCOPE

Farmaps lets users check reported medication prices and compare pharmacies across Bogotá. This version incorporates **security controls** and **incident management** to govern the service.

OBJECTIVE
Protect information and link every report to its review, remediation, and audit trail record.

- Confidentiality
- Integrity
- Availability

02 Four integrated modules

ITIL + SECURITY FUNCTIONAL ARCHITECTURE

P06 — Support & reports
Public support and incident reports without creating an account, with a unique tracking ID.
`/soporte` Public →

P07 — Security & privacy
Security posture, privacy policy, CIA triad, and responsible disclosure.
`/seguridad` Public →

A07 — Service desk
ITIL service desk and P1 to P4 ticket management with a 1-business-day SLA.
`/admin-soporte` Admin →

A08 — Audit log & SLAs
Immutable administrative audit log and SLA compliance tracking.
`/auditoria` Admin →

SERVICE WORKFLOW

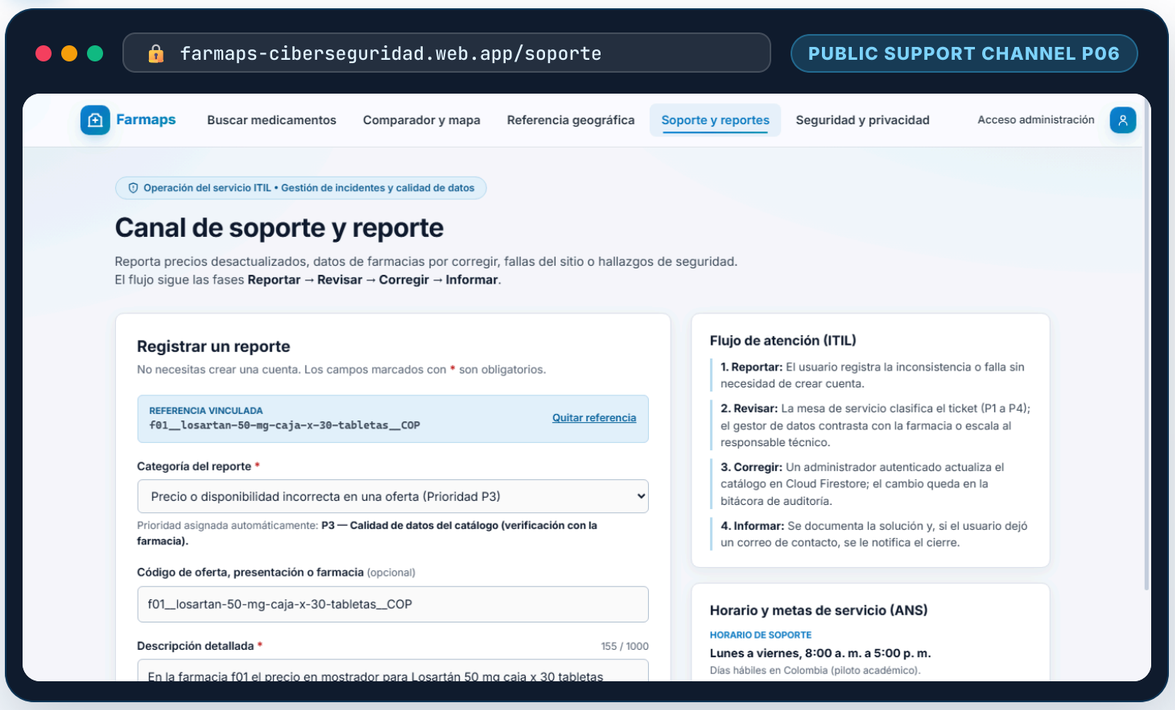
ITIL OPERATION

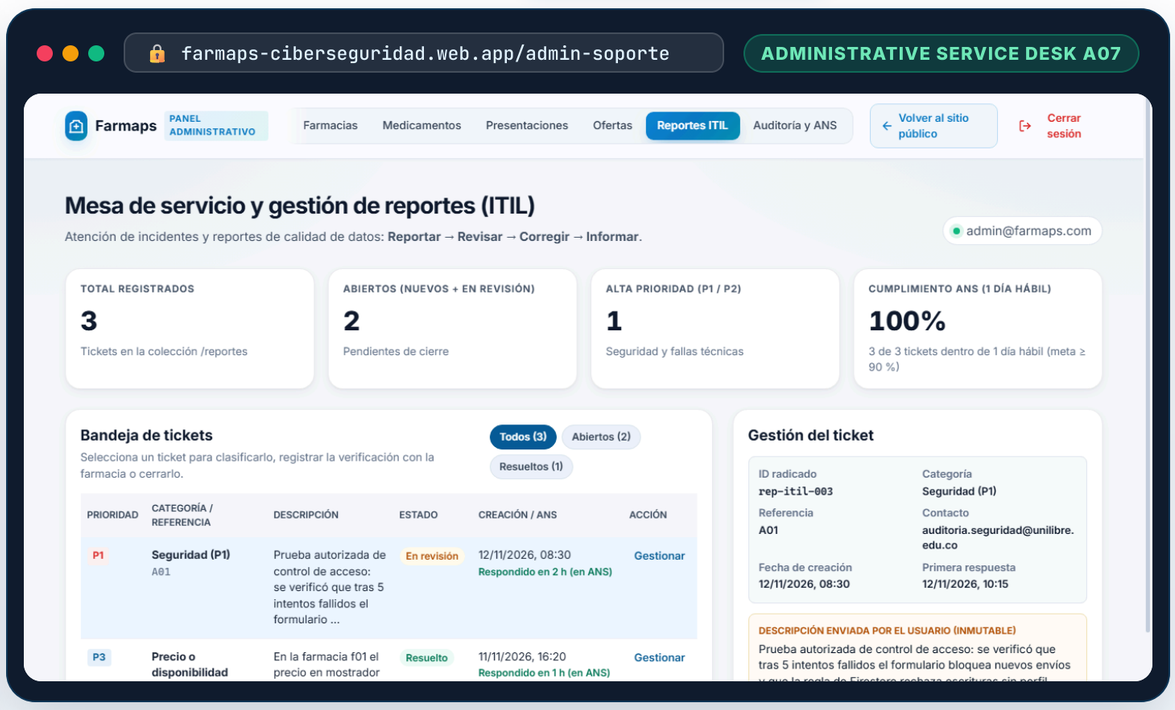
End-to-end incident traceability

1. **Report** — The user submits the incident.
2. **Review** — The admin verifies and prioritizes.
3. **Resolve** — The root cause or data is fixed.
4. **Inform** — The ticket is documented and closed.

Production implementation · farmaps-ciberseguridad.web.app

P06 Support + A07 ITIL Desk · P07 Security + A08 Audit Log





📌 Screenshots from the documented implementation. Administrative access requires authorization. (Click any screen to enlarge).

View P06 · View P07 · View A07 · View A08

03 Security controls

ACTIVE HARDENING

Access & session
Progressive local lockout after 5 failed attempts and automatic logout after 15 minutes of inactivity.
⏱ Max 5 attempts · 15m timeout

Permissions & traceability
Private reports and append-only audit log enforced through Cloud Firestore security rules.
🛡 Firestore Append-Only

Browser protection
HTTPS, HSTS, strict CSP, and active defense against clickjacking and MIME sniffing.
🔒 HSTS · CSP · X-Frame DENY

Responsible disclosure
Dedicated P1 critical finding channel and standardized `security.txt` file.
RFC 9116 /.well-known/

04 Management & verification

ITIL V3/V4 + ETHICAL HACKING

APPLIED ITIL LIFECYCLE — 5 PHASES

Strategy	Service value and scope
Design	Data, resources, and security
Transition	Testing and isolated deployment
Operation	Support and incident management
Continual imp.	Audit trail and indicators

ETHICAL HACKING — 6 SCENARIOS

The manual documents **6 permission test scenarios (S17 to S22)** to verify access control, report validation, and audit log integrity:

- S17 Private reads
- S18 P06 Anti-Spam
- S19 Catalog RBAC
- S20 Append-Only Log
- S21 Immutability
- S22 Traceability

PILOT TARGETS

Tracking objectives. They do not represent verified results on this poster.

≥ 99 %	≥ 90 %	≥ 90 %
Monthly availability	Offers updated within 7 days	First response in 1 business day

EXPLORE THE LIVE APPLICATION

farmaps-ciberseguridad.web.app

Informational service. Confirm price and stock directly with the pharmacy.

SCAN TO ENTER